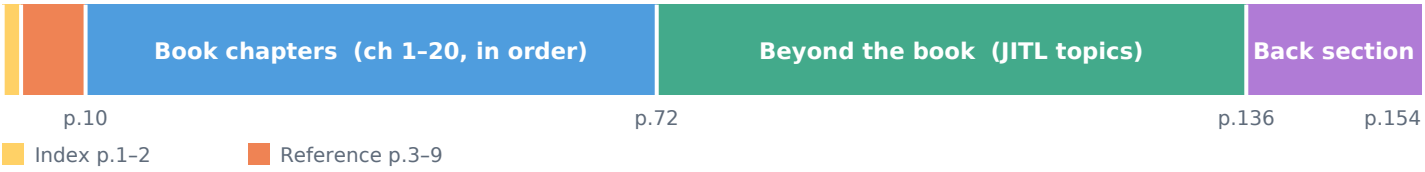
CCNA Notebook Guide

How to set up and fill the pen+GEAR 1-subject notebook (77 sheets = 154 pages, college ruled)
Method: Cornell notes · one topic per two-page spread · cues are questions · summary after the lab

1 Page map: decide where everything goes before you write



Pages	Section	What goes there
1–2	Index	Topic → page number. Fill it in as you go. (Mirror in the app: Book → Notebook.)
3–9	Reference	pp.3–8: subnetting, ports, AD/route codes, IOS modes, config cheat sheet, OSI chart (guide pages 5–8). p.9: your tab-colour key (guide page 8).
10–71	Book chapters	One spread per chapter you read or skim, starting on an even page: about 3–4 pages each, 6 for Ch 8 and Ch 20. Skip chapters 4, 14 and Appendix A.
72–135	Beyond the book	About 4 pages each: VLANs, STP, EtherChannel, OSPF, IPv6, FHRP, NAT, services, QoS, security, wireless, automation
136–154	Back section	Work backwards: mistake log from p.154, weekly review log from p.145 (guide page 9)

2 Setup tonight (≈20 min)

1. Write the page number in the corner box of every page, 1–154. Odd numbers are right-hand pages, even numbers left-hand.
2. Pages 1–2: copy the index from guide pages 4–5. Its page numbers are your budget.
3. Pages 3–9: copy the reference pages and tab key (guide pages 5–8).
4. Rule the first chapter spread (pages 10–11) as shown on page 2. Every spread starts on an even page.
5. Page 154: write MISTAKE LOG. Page 145: WEEKLY REVIEW.
6. Stick a tab on p.3 (Reference) and p.154 (Mistakes).
7. Don't use the perforations. Torn-out pages break the index.

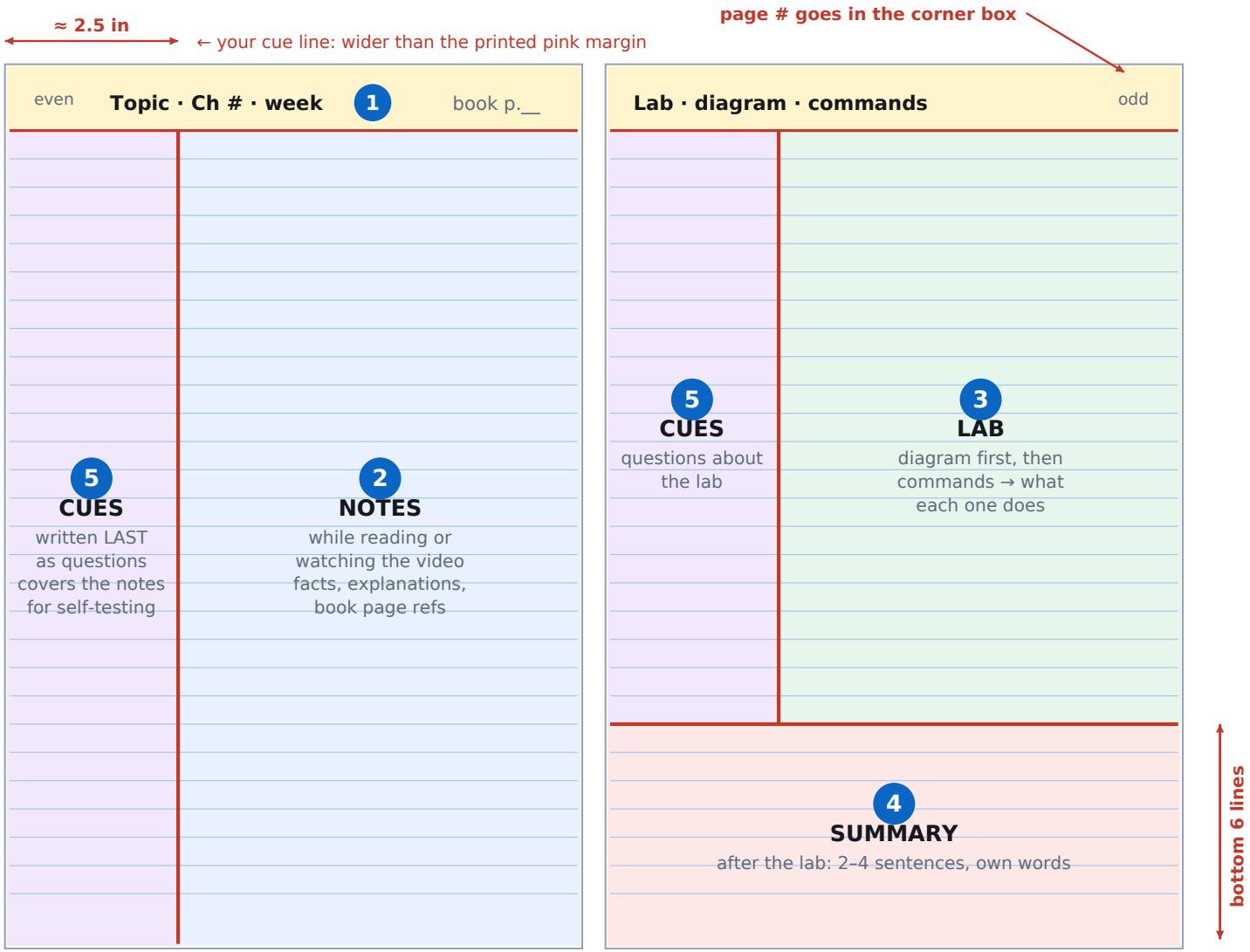
3 Sticky-tab colour key

One colour per exam domain. Flag every spread's right-hand page in its domain colour (side edge); section flags on the top edge. Colour each box with the flag you choose:

- ☐ Network Fundamentals — exam 20%
- ☐ Network Access — exam 20%
- ☐ IP Connectivity — exam 25%
- ☐ IP Services — exam 10%
- ☐ Security Fundamentals — exam 15%
- ☐ Automation & Programmability — exam 10%
- ☐ Reference pages
- ☐ Mistake log

How to rule each topic spread

Draw these lines in pen (red below) on every new topic. Each colour zone has one job, and you fill them in this order: 1 → 5.



The 5 zones

1 Header line
Topic, chapter, week and book page, written beside the corner box (the page number goes in the box). Lets the index point here.

2 Notes (left page)
While reading or watching. Short phrases, arrows, your own abbreviations. Leave gaps to add to later.

3 Lab (right page)
Draw the topology first. Then every command you ran, with what it does in plain words.

4 Summary
Write it only after the lab, not after the video. If you can't summarise it, redo the lab.

5 Cues (both pages)
Last step. Turn each block of notes into a question. Cover the notes and answer out loud.

— = lines you draw yourself (pen + ruler, or the edge of the book)

Big topic (e.g. OSPF)? Continue onto the next spread and keep the same layout. Only the last right page gets the summary box.

Worked example: Chapter 2, encapsulation

Blue = notes and lab. Purple = cues (written last). Red = the lines you drew. Your handwriting will be bigger: let it spill onto a second spread.

12

Ch 2 · Encapsulation · wk 1 *book p.__*

5 steps of encapsulation?

SENDING side: 5 conversion steps

1. Build data (app: e-mail text)

2. Package for transport → SEGMENT

```
TCP/UDP header + port numbers
```

Which layer adds src/dest IP?

3. Add network header → PACKET

```
src + dest IP → routers read this
```

Who adds MACs?

4. Add data-link header → FRAME

```
src + dest MAC + FCS trailer
```

Why rebuilt at every hop?

```
!! new frame at EVERY hop
```

5. Convert to BITS on the cable

PDU names L4 / L3 / L2 / L1?

PDUs: Data → Segment → Packet → Frame → Bits

"Don't Some People Fear Birthdays"

What happens on the receiver?

RECEIVING side = de-encapsulation

each layer strips its header,

passes the rest UP

OSI ↔ TCP/IP?

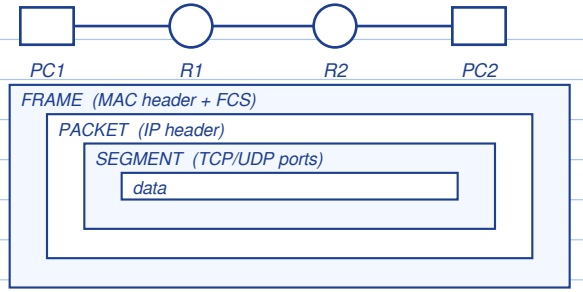
OSI 5-6-7 = TCP/IP Application

OSI 3 = Internet OSI 1-2 = Link

see also: Ch 9 (TCP), Ch 19 (ARP)

13

Lab: watch encapsulation hop by hop



Which addresses change at a router?

app Lab 5: ping PC1 → PC2

```
ping 192.168.2.10
→ watch it hop PC1-R1-R2-PC2
R1# show arp
  = MACs R1 learned (L2 addresses)
```

IP src/dst = SAME end to end

MAC src/dst = rewritten by EACH router

SUMMARY

Each layer wraps the one above in its own header (encapsulation).

IP addresses get the packet across networks and never change.

MAC addresses only get it across ONE link, so every router strips the frame and builds a new one. Receiver unwraps in reverse.

Index (p.1): written on the page

Page 1 is a right-hand page, so the index starts opposite the inside cover. The page numbers are a ready-made budget: copy them as-is.

inside front cover

Name: sin

CCNA 200-301 v1.1

Exam: Fri Jan 29, 2027

v1.1 last day: Feb 2

Book: CCNA 1 & 2 Companion Guide 3e

If found, please return to:

_____

INDEX *1*

Topic	Pages	Tab
BOOK CHAPTERS		
Ch 1 Binary, hex, IP & masks	*10–11*	*Fund*
Ch 2 OSI, TCP/IP, encapsulation	*12–15*	*Fund*
Ch 3 Media (skim)	*16–17*	*Fund*
Ch 5 Ethernet frames, MAC	*18–21*	*Fund*
Ch 6 Domains, switching, STP intro	*22–25*	*Access*
Ch 7 TCP/IP suite, IP classes	*26–29*	*Fund*
Ch 8 Routing basics + SUBNETTING	*30–35*	*Fund*
Ch 9 TCP/UDP, ports, apps	*36–39*	*Fund*
Ch 10 Inside a router (skim)	*40–41*	*IP*
Ch 11 IOS CLI basics (skim)	*42–43*	*IP*
Ch 12 Router configuration	*44–47*	*IP*
Ch 13 CDP, remote access	*48–49*	*Access*
Ch 15 Static routes, AD, protocols	*50–53*	*IP*
Ch 16 Reading the routing table	*54–55*	*IP*
Ch 17 ICMP, ping, traceroute	*56–57*	*Fund*
Ch 18 Troubleshooting	*58–61*	*IP*
Ch 19 TCP in depth, ARP	*62–65*	*Fund*
Ch 20 ACLs	*66–71*	*Sec*
skipped: Ch 4, Ch 14, App A (no spread)		
Tabs: Fund · Access · IP · Svc · Sec · Auto		

Index (p.2) and subnetting reference (p.3)

Red = lines you draw (column lines, header rule). Blue = your writing. Each row of text sits on one ruled line.

2

INDEX (cont.)

Topic	Pages	Tab
FRONT		
Reference: subnet 3 · ports 4 · AD 5	3–5	Ref
IOS 6 · configs 7 · OSI 8 · tab key 9	6–9	Ref
BEYOND THE BOOK		
VLANs & trunks	72–75	Access
Inter-VLAN routing	76–77	Access
STP / Rapid PVST+	78–81	Access
EtherChannel, LLDP	82–83	Access
OSPF	84–89	IP
IPv6	90–93	IP
FHRP / HSRP	94–95	IP
NAT & DHCP	96–99	Svc
DNS, NTP, SNMP, syslog, SSH	100–103	Svc
QoS	104–105	Svc
L2 security (port sec, snooping, DAI)	106–109	Sec
AAA, wireless security	110–111	Sec
Wireless architectures, WLC	112–115	Access
SDN, REST, JSON	116–119	Auto
Ansible/Terraform, AI	120–121	Auto
spare / overflow	122–135	
BACK (written backwards)		
Weekly review log	145 ←	
Mistake log	154 ←	Mist.

3

REF · Subnetting

```
Bits  128  64  32  16   8   4   2   1
Mask  128 192 224 240 248 252 254 255
```

Block = 256 – mask octet · Wildcard = 255 – mask octet

/	Mask	Block	Hosts
/16	255.255.0.0	1 (2nd)	65,534
/17	255.255.128.0	128 (3rd)	32,766
/18	255.255.192.0	64 (3rd)	16,382
/19	255.255.224.0	32 (3rd)	8,190
/20	255.255.240.0	16 (3rd)	4,094
/21	255.255.248.0	8 (3rd)	2,046
/22	255.255.252.0	4 (3rd)	1,022
/23	255.255.254.0	2 (3rd)	510
/24	255.255.255.0	1 (3rd)	254
/25	255.255.255.128	128 (4th)	126
/26	255.255.255.192	64 (4th)	62
/27	255.255.255.224	32 (4th)	30
/28	255.255.255.240	16 (4th)	14
/29	255.255.255.248	8 (4th)	6
/30	255.255.255.252	4 (4th)	2

Network = multiple of block ≤ host octet

Broadcast = next multiple – 1

Hosts = 2^h – 2 · Subnets = 2^n (subnet zero OK)

e.g. 172.16.45.200/27 → block 32 → net .192, bcast .223

Reference: ports (p.4) and AD / route codes (p.5)

Copy these once; add to them whenever Anki or a quiz catches you out.

4

REF · Port numbers

TCP	UDP
FTP 20 data / 21 ctrl	DNS 53
SSH 22	DHCP 67 srv / 68 client
Telnet 23	TFTP 69
SMTP 25	NTP 123
DNS 53 (zone transfer)	SNMP 161 / traps 162
HTTP 80	Syslog 514
POP3 110	RIP 520
IMAP 143	CAPWAP 5246 ctrl / 5247 data
BGP 179	RADIUS 1812 / 1813
HTTPS 443	
TACACS+ 49	

Well-known 0–1023 · registered 1024–49151

Ephemeral (client source port) 49152–65535

IP protocol #: ICMP 1 · TCP 6 · UDP 17 · OSPF 89

UDP = no handshake: DNS, DHCP, TFTP, SNMP, syslog, voice

5

REF · AD & route codes

Source	AD	Code	Meaning
Connected	0	C	connected subnet
Static	1	L	local /32 (own IP)
eBGP	20	S	static
EIGRP	90	S*	static default
OSPF	110	O	OSPF
RIP	120	D	EIGRP
EIGRP ext.	170	R	RIP
iBGP	200	B	BGP
Unusable	255	*	candidate default

CHOOSING A ROUTE

1) longest prefix match, always first

2) lowest AD (same prefix, different sources)

3) lowest metric (same protocol)

equal metric → load-balance (ECMP)

```
O 10.1.1.0/24 [110/2] via 10.0.12.2
```

[AD / metric] → AD 110 (OSPF), cost 2

Floating static = higher AD, e.g. 130 > OSPF 110

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1  → S*
```

Reference: IOS modes (p.6) and config cheat sheet (p.7)

p.6 = how to move around and what to check. p.7 = the command blocks you type in every lab (→ = enter that mode, then…).

6

REF · IOS modes

Prompt	Mode	In → Out
R1>	*user EXEC*	*login → exit*
R1#	*privileged*	*enable → disable*
R1(config)#	*global*	*conf t → end*
R1(config-if)#	*interface*	*int g0/0 → exit*
R1(config-line)#	*line*	*line vty 0 4 → exit*
R1(config-router)#	*routing*	*router ospf 1 → exit*

end / Ctrl-Z → straight back to R1#

do <command> → run a # command from config mode

MUST-KNOW SHOW COMMANDS

`sh ip int br`	*→ IP + status, 1 line each*
`sh run`	*→ active config (RAM)*
`sh start`	*→ saved config (NVRAM)*
`sh ip route`	*→ routing table*
`sh vlan brief`	*→ access ports per VLAN*
`sh int trunk`	*→ trunks + allowed VLANs*
`sh mac address-table`	*→ MAC → port*
`sh cdp nei (detail)`	*→ who is connected where*
`sh ip ospf nei`	*→ OSPF adjacencies*
`sh access-lists`	*→ ACL lines + hit counts*
`copy run start`	*→ SAVE IT!*

7

REF · Config cheat sheet

BASE + SSH

```
hostname R1 · enable secret ****
line con 0 → password **** · login
line vty 0 4 → login local · transport input ssh
username admin secret ****
ip domain-name lab.local · crypto key generate rsa
service password-encryption · banner motd #Keep out#
```

INTERFACES

```
int g0/0 → ip address 10.0.0.1 255.255.255.0 · no shut
SW: int vlan 1 → ip address … · ip default-gateway …
```

VLANS & TRUNKS

```
vlan 10 → name SALES
int f0/1 → switchport mode access · sw access vlan 10
int g0/1 → switchport mode trunk · sw trunk native vlan 99
```

ROUTER-ON-A-STICK

```
int g0/0.10 → encapsulation dot1Q 10
   ip address 10.0.10.1 255.255.255.0  (g0/0: no shut)
```

ROUTING

```
ip route 192.168.2.0 255.255.255.0 10.0.12.2
ip route 0.0.0.0 0.0.0.0 10.0.12.1   ← default (S*)
router ospf 1 → router-id 1.1.1.1
   network 10.0.0.0 0.255.255.255 area 0 · passive-int g0/2
```

ACLS

```
access-list 10 deny host 192.168.1.10
access-list 10 permit any → int g0/1 → ip access-group 10 out
ip access-list extended NAME → deny/permit … → apply in
copy run start   ← always save!
```

Reference: OSI chart (p.8) and tab key (p.9)

Colour the boxes on p.9 with the actual sticky flag you choose for each domain.

8

REF · OSI ↔ TCP/IP

#	OSI	TCP/IP	PDU	Device
7	Application	Application	Data	
6	Presentation	"	Data	
5	Session	"	Data	
4	Transport	Transport	Segment	firewall
3	Network	Internet	Packet	router
2	Data Link	Link	Frame	switch, AP
1	Physical	Link	Bits	hub, cable

PROTOCOLS / ADDRESSES

7 HTTP · DNS · FTP · SMTP · SNMP · SSH

6 encryption, compression, formats

5 session setup / teardown

4 TCP, UDP · port numbers

3 IP, ICMP, OSPF · IP addresses

2 Ethernet, 802.1Q · MAC addresses

1 UTP, fiber, radio · bits

All People Seem To Need Data Processing (7 → 1)

Don't Some People Fear Birthdays (PDUs):

Data · Segment · Packet · Frame · Bits

9

TAB KEY

Colour	Domain	Exam %
☐	Network Fundamentals	20%
☐	Network Access	20%
☐	IP Connectivity	25%
☐	IP Services	10%
☐	Security Fundamentals	15%
☐	Automation & Prog.	10%
☐	Reference pages	—
☐	Mistake log	—

WHERE THE FLAGS GO

Side edge: every spread, on its right-hand page, in its domain colour. Stagger them down the edge.

Top edge: sections. Index p.1 · Reference p.3 · Beyond the book p.72 · Back p.136

Weight = how much of the exam: IP Connectivity (25%) deserves the most flags and review.

Exam Fri Jan 29 · v1.1 last day Feb 2

The back section (pp. 136–154)

Both logs start at the back of the notebook and grow towards the middle, so they never collide with your spreads.

Mistake log: start on p.154 and work backwards

One line per missed question from Boson, the app's chapter quizzes or Anki. The app keeps its own Weak topics list automatically; this log is for the misses you want to think through. Three misses on the same topic → redo that spread from memory.

Date	Source	Topic	What I got wrong	The right idea	Spread
11/14	Quiz ch 8	Subnetting	Said 6 subnets for 3 borrowed bits	$2^3 = 8$ (subnet zero is allowed)	p.30
11/16	Anki	Ports	TFTP = TCP 69	TFTP = UDP 69 (no handshake)	p.4

Weekly review log: start on p.145 and work backwards

Every Sunday at 4 pm, one row per week. Compare the hours with the plan's split (60/40 theory/practice in weeks 1–2, 40/60 after that, 20/80 in week 12).

Wk	Hours (theory / practice)	Chapters / topics done	Lab redone from memory	Weakest topic → next week
1	5 / 4	Ch 1, 2, 5	Lab 1 Hello, switch	hex conversions
2				
3				

Using the notebook every session

One study session, in order



Daily, 7:30 pm (10–15 min)

- Anki deck, then 10 min of subnetting (app → Drills).
- Glance at today's spread and answer the cues out loud.
- Log any misses in the back of the notebook.

Sunday, 4 pm weekly review

- Cover the notes column on every spread from this week and answer the cues.
- Redo one lab from memory, without the brief. Check it with the app's Check button.
- Fill one weekly-review row (p.145 backwards).
- Any cue you missed: rewrite that bit of the notes, or redo the spread.

Rules of thumb

- Facts go in Anki (port numbers, AD values). How things work goes in the notebook.
- Cues are questions, not facts: "Why does the frame change at each hop?", not "frames change".
- Diagrams beat paragraphs. Draw the topology for every lab.
- Commands always get a meaning: "no shutdown → bring the interface up".
- Leave a blank page at the end of each domain for later additions.
- Book chapters you skip (4, 14, App A) get no spread; their one-page summaries are in the app.
- Annotate the book lightly: mark skip/skim and write the notebook page in the margin.

How the notebook and the app fit together

- Book tab → chapter: tick the reading guide, set the Book page and the Notebook page, take the chapter quiz.
- Book → Notebook: the digital copy of pages 1–2, with page numbers for every chapter and beyond-the-book topic.
- Cornell notes in the app = your cue questions, so you can self-test on the phone. Paper keeps the diagrams.
- Quiz misses land in Today → Weak topics automatically; copy the ones you need to think about into the mistake log.